

WEN XIE

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Ph.D. applying ML/AI to business problems at scale. 6+ years in computer vision, NLP, and causal inference. Built end-to-end pipelines analyzing 100M+ records at Apple and Snap. Led project awarded Amazon Research Award. Best conference paper award winner. Published in *J. of Advertising*, *J. of Advertising Research*, *ICPR*, *Int. Conf. on Multimedia Modeling*.

SKILLS

Programming & Tools: Python, SQL, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter

ML/DL/CV/NLP: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, OpenCV, YOLO, Object Detection, Image Segmentation, BERT, T5, BART, Summarization, Question Answering, Agentic AI, Recommendation System

Statistics & Causal Inference: A/B Testing, Propensity Score Matching, Instrumental Variables, Regression Models

EXPERIENCE

Postdoctoral Research Fellow

Feb 2024 - Present *Northeastern University, Institute for Experiential AI*

- Developed video summarization pipeline for ad shortening, achieving 0.783 AP using multi-modality fusion
- Applied video embedding + clustering to match content of short and long ads at shot-level on 60K+ video ads; found that short TV ads attract 21% more attention during viewer presence; long TV ads receive 5% more purchase intent
- Built large-scale ad-response prediction models (XGBoost, Two-Tower, DLRM) using viewer/ad ID embeddings, categorical and contextual features, and deep feature fusion, achieving 0.83 AUC and 0.89 PR-AUC on 30M+ TV ad impressions
- Led 4+ ML projects in CV/NLP for marketing analytics, analyzing 60M+ TV impressions (TVision) and 6M+ reviews (Yelp), deploying OpenAI APIs for feature extraction, conducting hypothesis tests, and contributing to advertisers

Research Scientist Intern, Computational Social Science Team

Summer 2022 *Snap Inc.*

- Analyzed 50M+ user records using SQL and causal inference to measure content-ad congruence impact on engagement
- Built data pipeline: engineered 30+ features, implemented propensity score matching to address selection bias
- Discovered 11% lift in ad viewing time from content-ad alignment; finding led to A/B test launch for ad ranking

Machine Learning Research Intern, Web Answers and Ranking Team

Summer 2021 *Apple Inc.*

- Designed multi-hop QA system using Dense Passage Retrieval + text summarization + generative models (BART, T5)
- Fine-tuned models on 400K+ QA pairs, improving Exact Match by 30% and F1 by 20%; reduced input tokens by 60%
- Conducted ablation studies comparing extractive vs. abstractive summarization approaches for production team

EDUCATION

Ph.D., Electrical Engineering | University of Houston | GPA: 3.85/4.0

2018 - 2023

B.Eng., Electronic Information Engineering & B.Econ., Finance | UESTC | GPA: 3.95/4.0

2014 - 2018

SELECTED RESEARCH PROJECTS

Automated Skin-Tone Diversity Detection | *Published at HICSS 2023, under review at Journal of Marketing*

- Developed CV pipeline using Swin Transformer (86% mIOU) to quantify diversity metrics from 70K+ brand images/videos
- Applied Bayesian change-point detection to identify post-2020 strategy shifts; found diversity-alignment between brand models and their audiences drives 15% engagement lift

Eye-Tracking Analysis with Object Detection | *Amazon Research Award 2021, Published in Journal of Advertising 2024*

- Conducted eye-tracking experiments with recruited participants shopping online and on mobile devices
- Trained YOLO to extract ad elements from 235 videos (94.5% mAP); reduced annotation time by 95%
- Built regression models revealing mobile ads receive 40% less attention with users focusing 2.3x more on text vs. images

ENTREPRENEURIAL PROJECTS

Co-Founder & AI Engineer | *Immigration AI Agent*

2025 - Present

- Built agentic AI system automating immigration document filling and generation using LLMs and RAG-based retrieval
- Developed multi-agent workflow: web crawling for customer needs, document processing, and PDF generation
- Architected full-stack platform serving paying customers; integrated LLM APIs, database, and user-friendly interface

Co-Founder & Developer | *Maison Carina*

2025 - Present

- Built e-commerce platform for handcrafts from my mom; deployed end-to-end web solution for payments, inventory